Problem Set: 2
Computer Networks and the Internet: Application layer

1 Essay Questions

1. Suppose you wanted to do a transaction from a remote client to a server as fast as possible. Would you use UDP or TCP? **Why?**

2. For a P2P file-sharing application, do you agree with the statement, "There is no notion of client and server sides of a communication session"? Why or why not?

3. List the four broad classes of services that a transport protocol can provide. For each of the service classes, indicate if either UDP or TCP (or both) provides such a service.

4. Consider a new peer Alice that joins BitTorrent without possessing any chunks. Without any chunks, she cannot become a top-four uploader for any of the other peers, since she has nothing to upload. How then will Alice get her first chunk?

5. Consider your HTTP programming assignment was to be made over UDP not TCP/IP, how many sockets did the server have to handle?

6. For the client-server application over TCP, why must the server program be executed before the client program? For the client-server application over UDP, why may the client program be executed before the server program?

7. What is meant by "in band" and "out of band" protocols? which of those protocols are in band and which are out band: HTTP, DNS, TCP, SMTP, FTP?

8. Describe how Web caching can reduce the delay in receiving a requested object. Will Web caching reduce the delay for all objects requested by a user or for only some of the objects? Why?

2 Problems

9. Consider a short, 10-meter link, over which a sender can transmit at a rate of 150 bits/sec in both directions. Suppose that packets containing data are 100,000 bits long, and packets containing only control (e.g., ACK or handshaking) are 200 bits long. Assume that N parallel connections each get 1/N of the link bandwidth. Now consider the HTTP protocol, and suppose that each downloaded object is 100 Kbits long, and that the initial downloaded object contains 10 referenced objects from the same sender. Would parallel downloads via parallel instances of non-persistent HTTP make sense in this case? Now consider persistent HTTP. Do you expect significant gains over the non-persistent case? Justify and explain your answer.

10. Consider distributing a file of F = 15 Gbits to N peers. The server has an upload rate of u_s = 30 Mbps, and each peer has a download rate of d_i = 2 Mbps and an upload rate of u. For N = 10, 100, and 1,000 and u = 300 Kbps, 700 Kbps, and 2 Mbps,

prepare a chart giving the minimum distribution time for each of the combinations of N and u for both client-server distribution and P2P distribution.

11. Consider the networks shown in the figure below. There are two user machines m1.a.com and m2.a.com in the network a.com. Suppose the user at m1.a.com types in the URL www.b.com/bigfile.html into a browser to retrieve a 1Gbit (1000 Mbit) file from www.b.com. List the sequence of DNS and HTTP message sent/received by/from m1.a.com from the point the URL is entered into the browser until the file is completely received. Indicate the sequence of messages as well as the source and destination of each message. You can assume that every HTTP request by m1.a.com is first directed to the HTTP cache in a.com and that the cache is initially empty, and that all DNS requests are iterated queries.

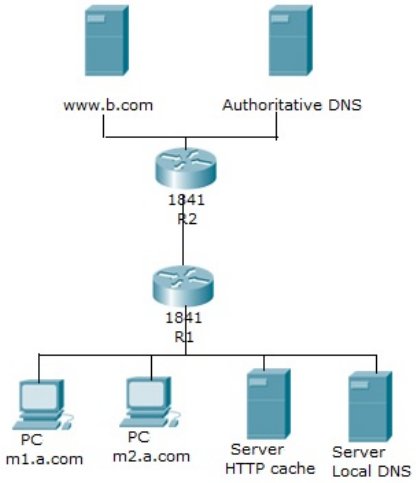


Figure 1: problem 3

3 Notes

- Your answers should be clear, concise and to the point.
- It's encouraged that you take a look at the Homework Problems for chapter 2, from the kurose/Ross reference.

Good Luck